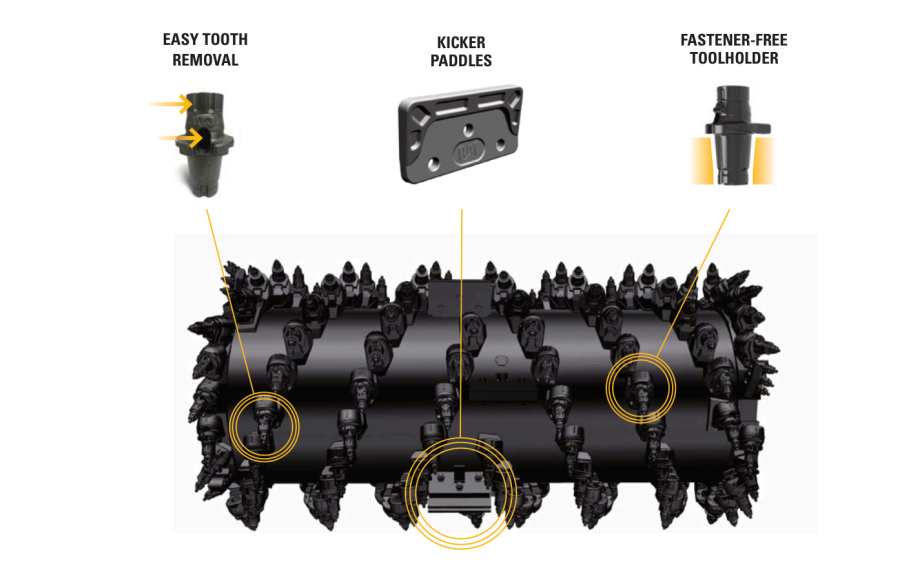
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Milling and reclaiming

Volume 5 · H1, H2, H3, H4, H5, H6, H7, H8

Classroom development edition · 30 October 2026



Caterpillar. (2026). Cold Planer Parts Reference Guide (QEXQ3131-01 (05/2026), source PDF p. 8). Faithful crop; original labels retained. Cat System K rotor example. Other rotor systems and retaining arrangements differ; no change-out procedure is taught here.

In this volume

1. How to use the volume and key terms
2. Milling and reclaiming equipment
3. Milling and reclaiming safety procedures
4. Milling and reclaiming operator roles and responsibilities
5. Milling and reclaiming equipment maintenance
6. Markers, grades and stakes
7. Operate milling and reclaiming equipment
8. Follow shut down procedures
9. Transport milling and reclaiming equipment
10. Hand over a controlled milling pass

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for milling and reclaiming. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS
Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT
Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline H1

Milling and reclaiming equipment

Milling removes pavement to a controlled profile. Reclaiming processes existing pavement and underlying material to form a designed layer. Understanding the material route helps a learner recognize the machine's working parts and the people who must stay clear of them.

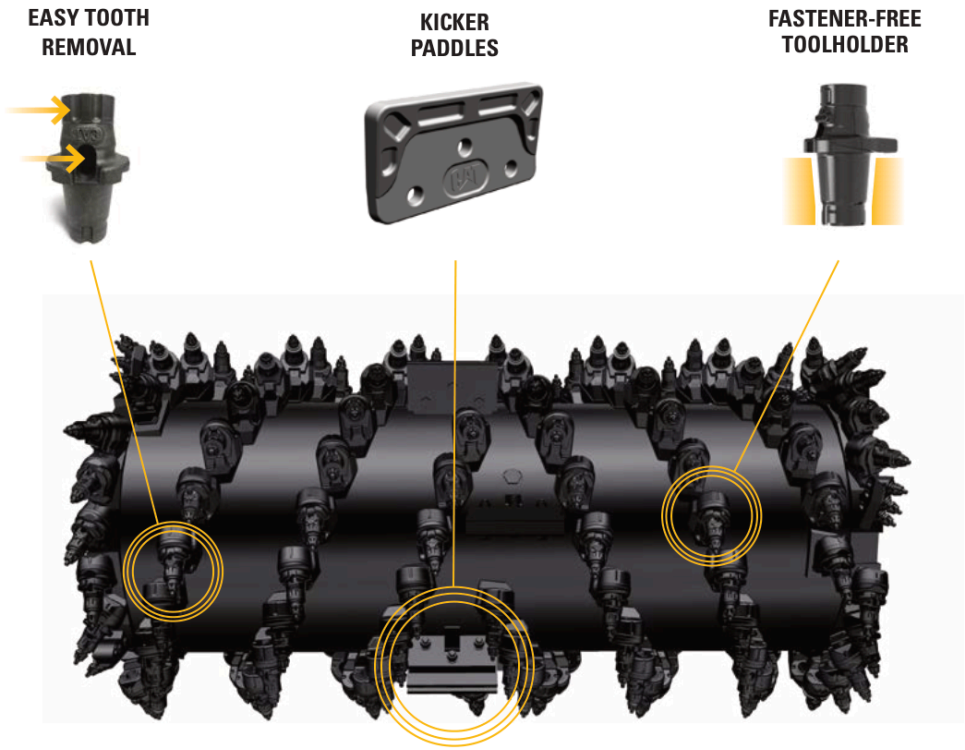
- Milling equipment
- The function of milling equipment
- Major mechanical components
- Potential failures, symptoms and indicators of failure

LEARNING STEP 1

Milling equipment

Identify manufacturer, model, serial number, rotor configuration and fitted options before selecting information. The CAT cold-planer parts guide illustrates assemblies, while the performance handbook explains equipment families. Neither replaces the matching operation and maintenance manual (OMM). A brochure can show a machine's purpose without describing its safe operating sequence. Record which document supports identification, which supports operation and which is specific to the training machine.

Cutting system: rotor details



Note. Caterpillar. (2026). Cold Planer Parts Reference Guide (QEXQ3131-01 (05/2026), source PDF p. 8). Faithful crop; original labels retained. Cat System K rotor example. Other rotor systems and retaining arrangements differ; no change-out procedure is taught here. Full source context.

LEARNING STEP 2

The function of milling equipment

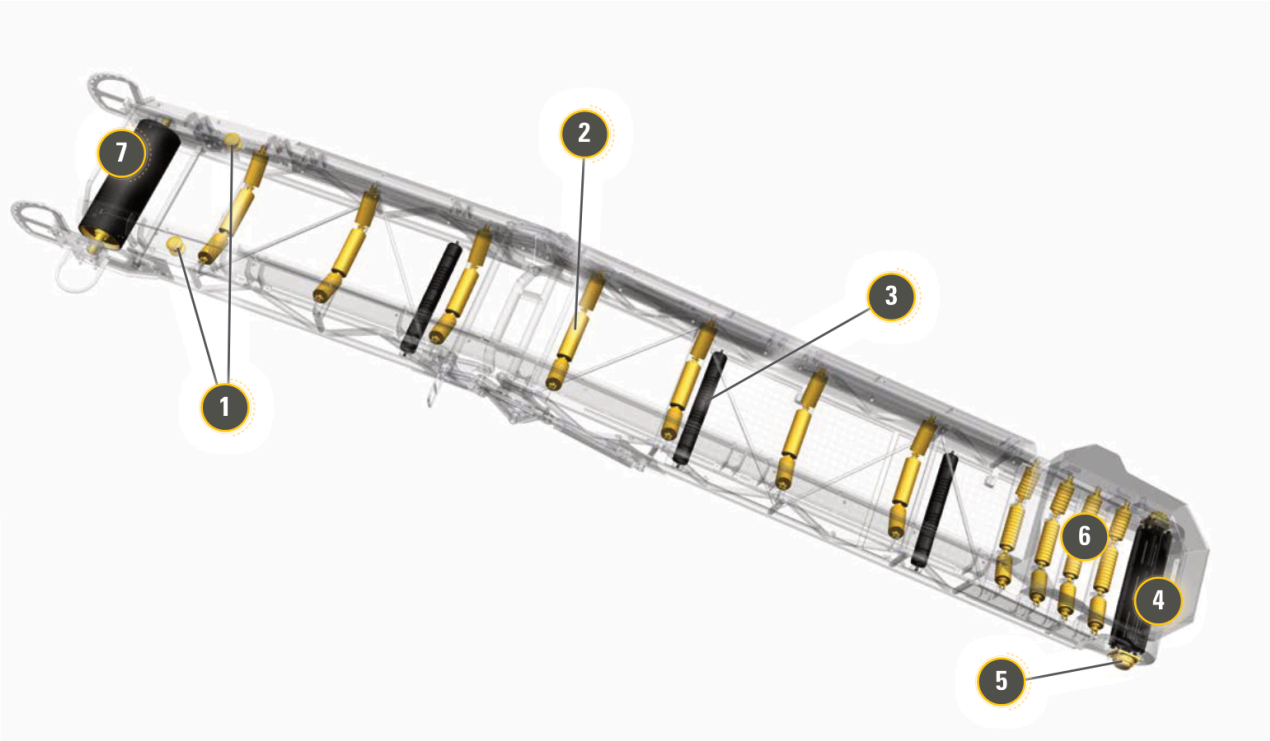
A cold planer uses a cutting rotor to remove a controlled depth, producing a milled profile and material for collection. A reclaimer mixes a designed depth of existing layers; specified water, binder or stabilizing agents may be introduced through a controlled system. Foamed asphalt is a particular binder-treatment process, not simply any hot asphalt mixed with water. Reclaiming, grading, compaction and paving form a coordinated sequence with design and testing requirements. Reuse can reduce virgin-material demand, but mixture suitability and performance still need verification.

LEARNING STEP 3

Major mechanical components

Trace power from engine through the drive systems to the rotor and travel system. Identify cutting tools and holders, rotor drum, cutting chamber, side plates, collecting and discharge conveyors, tracks or wheels, lifting columns, water/dust-control system, operator station, grade sensors and guards. The cutting tools fracture the surface; conveyors move the recovered material; grade controls influence the cut relative to a reference. Call a component by its actual manual name and point to it from a safe position. A guard is part of the machine's protection, not an optional cover.

Every numbered component in this conveyor diagram



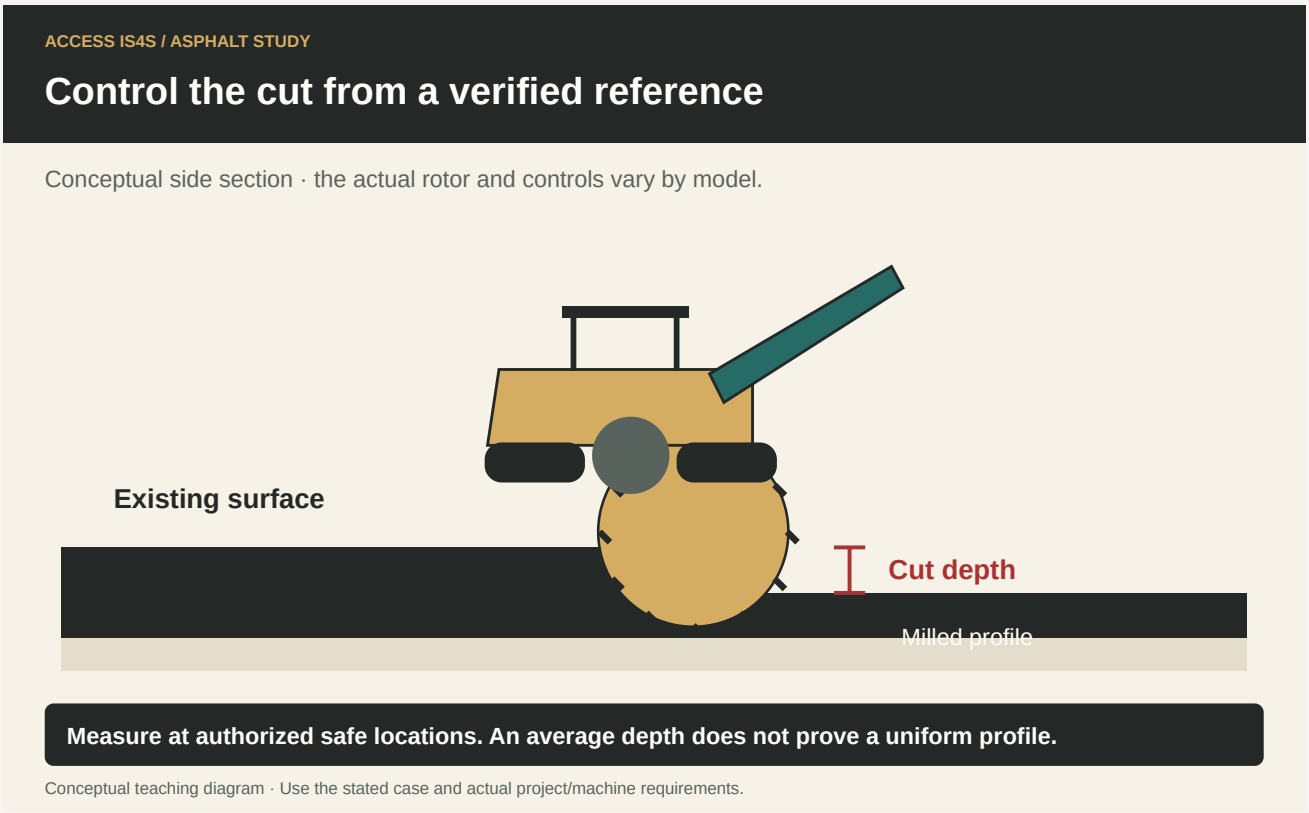
Note. Caterpillar. (2026). Cold Planer Parts Reference Guide (QEXQ3131-01 (05/2026), source PDF p. 25). Faithful crop; original labels retained. PM310, PM312, PM313, PM620, PM622, PM820, PM822 and PM825; individual parts/model tables apply. Full source context.

LEARNING STEP 4

Potential failures, symptoms and indicators of failure

Unusual noise, vibration, smell, leaks, damaged tools, cracks or warning messages are observations requiring action. State where and when they occur and whether conditions are worsening. Stop and secure the machine when a hazard or serious fault is suspected, keep others clear and notify the responsible person. Only authorized personnel diagnose or repair within their competence. Never search for a high-pressure leak with a hand or reach into a stopped-looking conveyor. Record the defect and authorized return-to-service decision.

Control the cut from a verified reference



Note. ACCESS IS4S. (2026). Control the cut from a verified reference [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A noise with a location

A new rhythmic noise and vibration begin at the discharge conveyor after a load change. A learner suggests increasing speed to clear it.

Think it through.

Report the change and location, stop safely under the machine procedure and protect the area. Have the responsible person arrange inspection under the required isolation. Increasing speed is an unsupported intervention and may worsen the fault.

Try it and explain your thinking

Match rotor, discharge conveyor, lifting column and grade sensor to their functions.

Match rotor, discharge conveyor, lifting column and grade sensor to their functions.

Skills you use: Reading, Communication.

A 90 m × 2 m strip is milled 0.04 m deep

A 90 m × 2 m strip is milled 0.04 m deep. Find its geometric removed volume. Is this its loose truck volume?

Skills you use: Numeracy, Problem solving.

Make a digital component card with model, image, name, function and hazard.

Make a digital component card with model, image, name, function and hazard.

Skills you use: Digital, Writing, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify ten components and explain their functions.	0 / 1 / 2	Evidence/help: _____
Distinguish milling, reclaiming and stabilized-base processes.	0 / 1 / 2	Evidence/help: _____
Record a fault and the appropriate hold/escalation.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, H1, printed p. 58; source PDF pp. 60–60.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 13: Rehabilitation Methods for Asphalt Pavements; source PDF pp. 653–710. Open publication.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- National Institute for Occupational Safety and Health. (2015; Publication 2015-105). Best Practice Engineering Control Guidelines to Control Worker Exposure to Respirable Crystalline Silica during Asphalt Pavement Milling. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline H2

Milling and reclaiming safety procedures

A milling plan must account for the cutting system, material discharge, truck movement, traffic and people on the ground. Agree on boundaries and communication before the rotor or travel system moves.

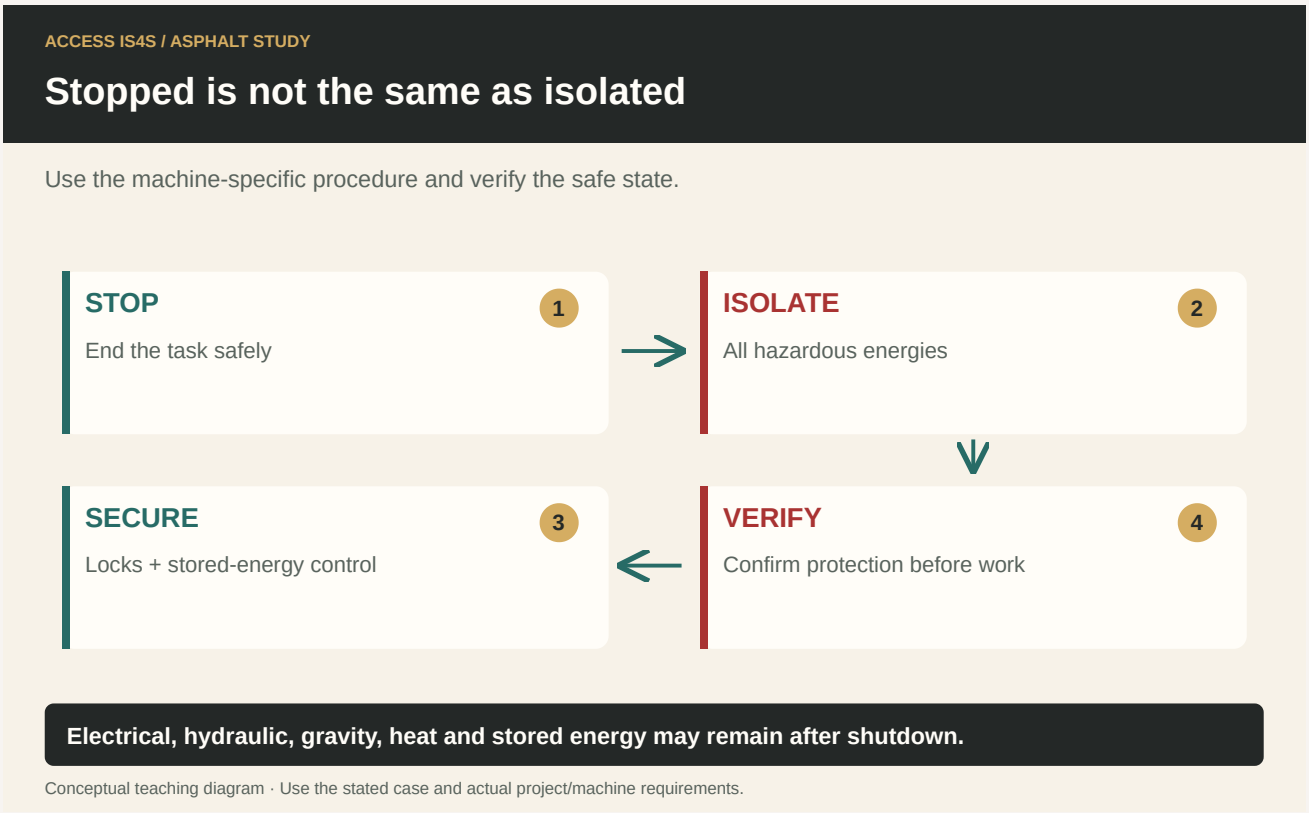
- Interpret and follow job plan
- Awareness of machine-specific safety
- Traffic Control requirements
- Proper PPE
- Lockout procedures
- Secure equipment for safety

LEARNING STEP 1

Interpret and follow job plan

Read the path, pattern, depth, pass sequence, utilities, obstacles, discharge arrangement and planned stopping points. Confirm the current revision and who can authorize a change. Walk the area only when access is controlled. Resolve utility information and unidentified obstructions before cutting. Brief the plan with ground crew, truck drivers and traffic-control personnel, using a drawing and read-back.

Stopped is not the same as isolated



Note. ACCESS IS4S. (2026). Stopped is not the same as isolated [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Awareness of machine-specific safety

Recognize rotor and conveyor entanglement, ejected material, silica-containing dust, exhaust, noise, hot surfaces, pressurized hydraulics, blind areas and raised components. Dust control is an engineered system requiring inspection and effective operation; a visible spray does not prove adequate exposure control. Block or support raised equipment using specified devices before access. Guard removal, clearing jams and service require the applicable isolation procedure.

LEARNING STEP 3

Traffic Control requirements

The traffic-control plan and site assessment determine TCP positions, numbers, signs, devices and communication. Milling traffic, public traffic and pedestrians can change relative positions as work advances. Keep a defined route and reassess driveways, intersections and service access. Only trained and authorized people perform assigned traffic-control roles. Stop if control or communication is lost.

LEARNING STEP 4

Proper PPE

Select PPE through the hazard assessment and current requirements, including suitable footwear, head, eye, hearing and high-visibility protection as applicable. Respiratory protection requires the employer's program, correct selection, training and fit testing where required. WorkSafeBC Part 8 assigns workers specified basic clothing, general-purpose gloves, footwear and safety headgear, while employers provide other required PPE; agreements may provide more. The employer must ensure required PPE is used and maintained.

LEARNING STEP 5

Lockout procedures

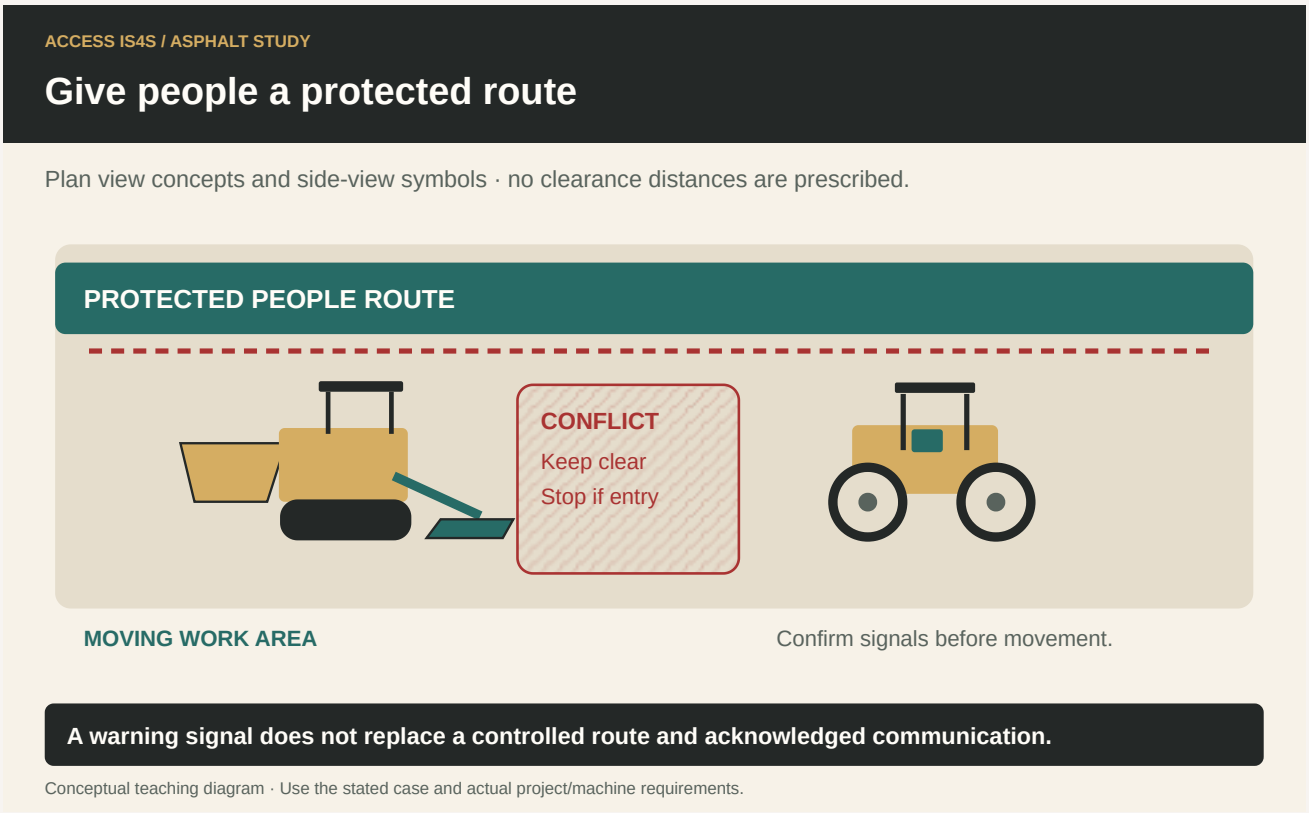
Identify every hazardous energy source, including engine/fuel, electrical, hydraulic, gravity, rotating parts and stored pressure. Use the machine-specific written lockout procedure: stop, isolate, secure, dissipate or restrain stored energy and verify the safe state before work. A key removed or battery switch off does not alone establish full isolation. Each worker's protection follows the applicable lockout system.

LEARNING STEP 6

Secure equipment for safety

Park in the designated stable location, lower or support components as instructed, apply parking controls and required chocks, and secure keys and access. Protect the public from the machine and residual hazards. Remove or secure detachable items without leaving guards missing. Record defects and communicate the machine's status to the next shift.

Give people a protected route



Note. ACCESS IS4S. (2026). Give people a protected route [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A changed discharge route

The planned truck position is blocked by a delivery vehicle. Someone proposes swinging the conveyor over a pedestrian route.

Think it through.

Stop the conflicting movements, preserve the pedestrian boundary and ask the lead to revise the arrangement with traffic control. Check overhead and swept-path hazards. Resume only after the new plan is communicated and controls are in place.

Try it and explain your thinking

Name four hazardous energies that can remain after engine shutdown.

Name four hazardous energies that can remain after engine shutdown.

Skills you use: Reading, Problem solving.

Write a read-back for a revised milling boundary at station 0+120.

Write a read-back for a revised milling boundary at station 0+120.

Skills you use: Writing, Communication, Collaboration.

A dust-control alarm appears while cutting

A dust-control alarm appears while cutting. What evidence should be retained?

Skills you use: Digital, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Brief and verify the work-zone boundary.	0 / 1 / 2	Evidence/help: _____
Demonstrate the approved lockout with all energy sources controlled.	0 / 1 / 2	Evidence/help: _____
Secure the parked machine and communicate its status.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, H2, printed p. 59; source PDF pp. 61–61.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 6: Substance Specific Requirements. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- National Institute for Occupational Safety and Health. (2015; Publication 2015-105). Best Practice Engineering Control Guidelines to Control Worker Exposure to Respirable Crystalline Silica during Asphalt Pavement Milling. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline H3

Milling and reclaiming operator roles and responsibilities

Milling is team work with distinct control responsibilities. Clear roles prevent two people giving incompatible instructions and ensure the operator receives reliable information about the cut and surrounding movement.

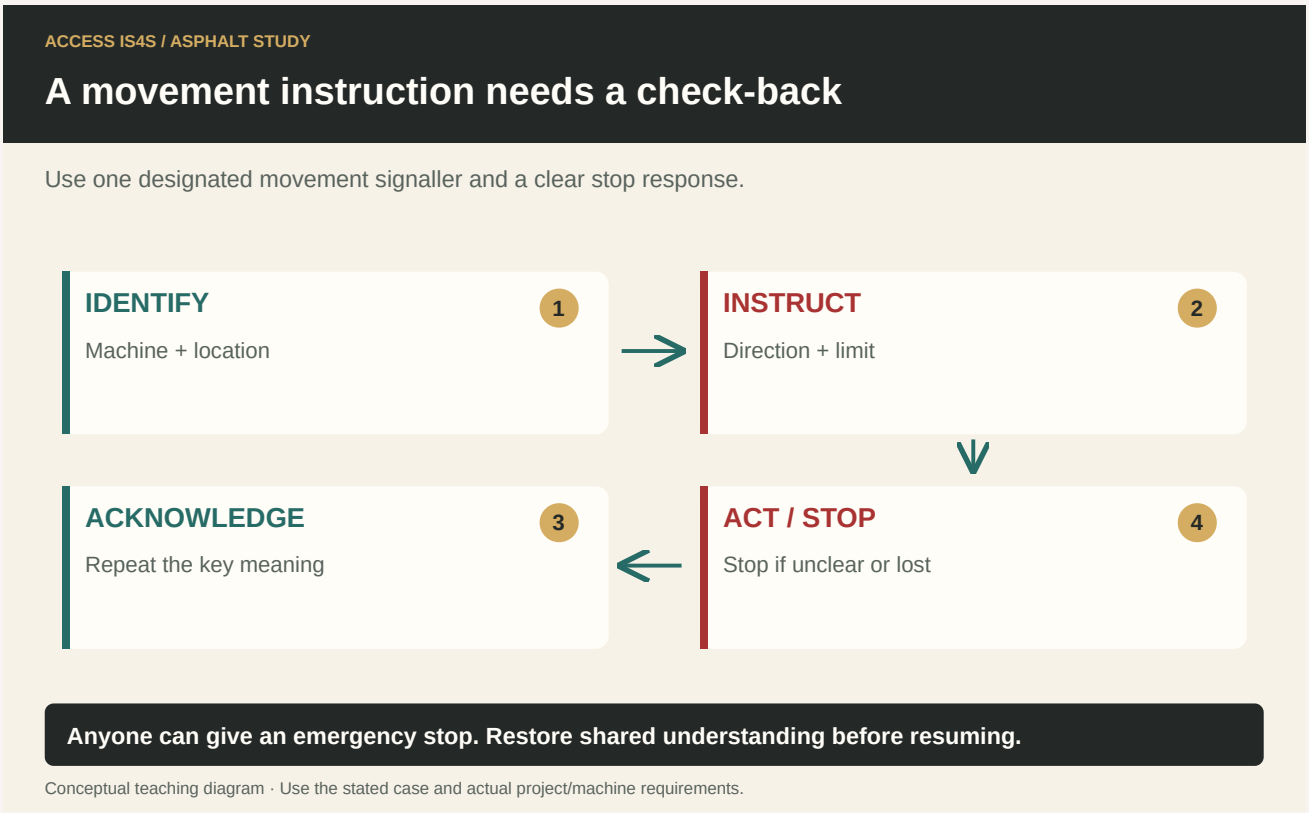
- The role of Milling Machine Operator
- The role of ground men
- Hand signals communication

LEARNING STEP 1

The role of Milling Machine Operator

The machine operator controls assigned travel and working functions from the operator station, monitors warnings and machine response, and maintains awareness of the work area. “Top end” describes a crew role, not permission to ignore ground observations. The operator must know who gives movement instructions, what the stop signal means and when to stop for uncertainty. A ground guide cannot remove the operator’s duty to operate safely.

A movement instruction needs a check-back



Note. ACCESS IS4S. (2026). A movement instruction needs a check-back [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

The role of ground men

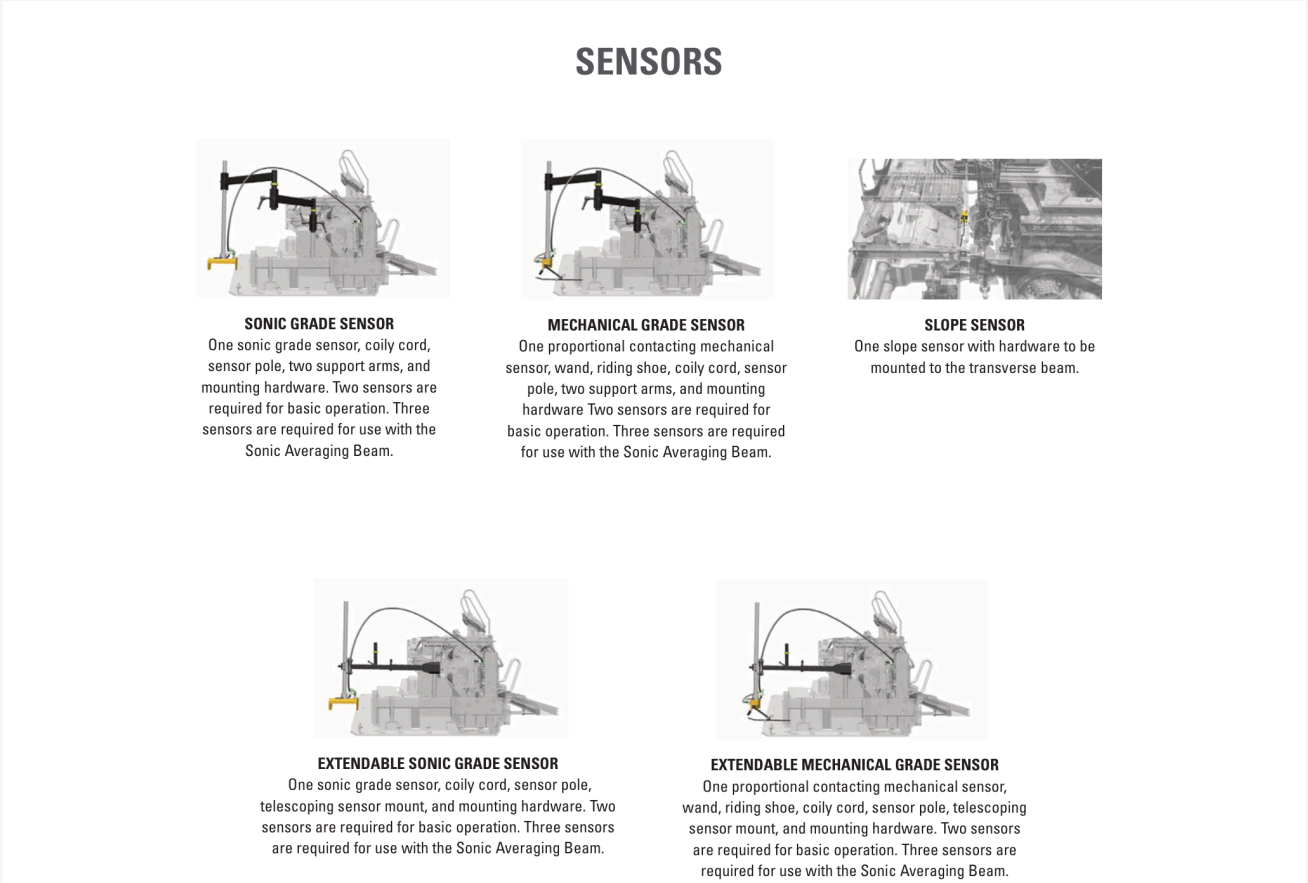
Ground crew members may check cut depth/profile, manage assigned grade controls, observe the material route and coordinate truck positioning from approved locations. Duties depend on the machine and work plan. No person should measure beside moving cutting equipment or stand in a truck's pinch zone to improve visibility. Agree who reports depth, who communicates with trucks and who records quality. Rotate or adjust work when fatigue, noise or visibility makes the assigned role ineffective.

LEARNING STEP 3

Hand signals communication

Before work, rehearse a small agreed set of movement and stop signals, radio wording and acknowledgement. Use one designated movement signaller at a time, while anyone can give an emergency stop. If a signal is unclear, conflicting or lost, stop and re-establish communication. Crane/hoisting work has applicable standardized signal requirements; a crew-created milling signal must not replace them. Use plain words with machine ID, direction and limits; avoid "over there" or an unexplained horn pattern.

Identify grade and slope sensors



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 34). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

Worked example · Two people signal

One ground worker signals a truck forward while another signals stop. The operator sees both.

Think it through.

Stop the affected movement. Clarify the hazard and designated signaller, confirm the driver and operator understand the revised instruction, and resume only when the area is clear and the communication arrangement is restored.

Try it and explain your thinking

Assign operator, depth checker, truck coordinator and recorder duties for a simulated pass.

Assign operator, depth checker, truck coordinator and recorder duties for a simulated pass.

Skills you use: Collaboration, Writing.

Depth readings at three authorized checks are 38, 41 and 40 mm against a fictional target 40 mm

Depth readings at three authorized checks are 38, 41 and 40 mm against a fictional target 40 mm. Find the mean and explain its limitation.

Skills you use: Numeracy, Problem solving.

Rehearse a lost-radio response with a partner, then improve the wording.

Rehearse a lost-radio response with a partner, then improve the wording.

Skills you use: Communication, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Perform a clear role and communication briefing.	0 / 1 / 2	Evidence/help: _____
Demonstrate stop on ambiguous or lost communication.	0 / 1 / 2	Evidence/help: _____
Report cut observations with location and units.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, H3, printed p. 60; source PDF pp. 62–62.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 3: Rights and Responsibilities. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 15: Rigging. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline H4

Milling and reclaiming equipment maintenance

Daily maintenance protects the next operator as well as the machine. Use the actual model's maintenance schedule and safe work instructions. Record what was checked, what was serviced and what remains out of service.

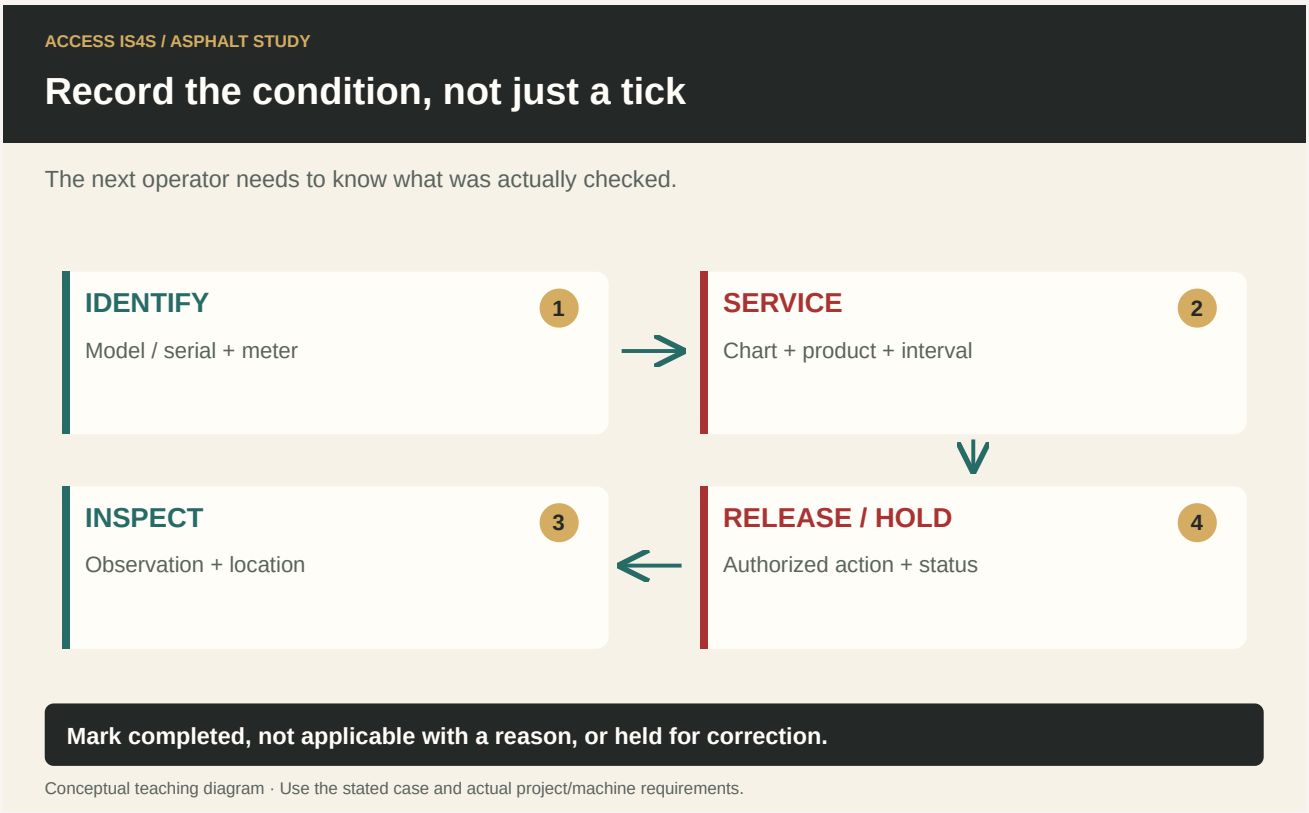
- Service lubrication system (greasing)
- Service air intake system as required (air filters / pre cleaners)
- Drain fuel tank sump, water separator fuel filters if equipped
- Add fluids as required
- Inspect and clean components as required
- Housekeeping tasks

LEARNING STEP 1

Service lubrication system (greasing)

Identify grease points from the machine chart, lubricant specification, interval and required machine position. Clean fittings before connection and load the grease gun according to its instructions. Use the specified quantity or completion criterion; generic numbers of gun strokes and visible overflow are not universal rules. Some bearings and centralized systems have different requirements. Do not reach past guards or into a hazard to access a fitting. Collect excess lubricant and report damaged fittings or blocked lines.

Record the condition, not just a tick



Note. ACCESS IS4S. (2026). Record the condition, not just a tick [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Service air intake system as required (air filters / pre cleaners)

Check the intake restriction indicator and service the pre-cleaner as instructed. Keep dirt out of the clean side of the system. Inspect seals and housings; a damaged seal can defeat a clean filter. Clean an element only where the manufacturer permits it and with its method and limits; otherwise replace it. Reduced-pressure air is not automatic permission to blow out every filter. Do not run without the required element.

LEARNING STEP 3

Drain fuel tank sump, water separator fuel filters if equipped

Where fitted, service the fuel sump/water separator using the specified safe conditions and collection method. Identify water, debris or unusual contamination and report it. Keep ignition sources controlled and avoid skin contact. Dispose of collected fluid through the approved route. Opening a high-pressure fuel line is not a routine water-drain task.

LEARNING STEP 4

Add fluids as required

Identify each reservoir and correct fluid before adding anything. Check on the specified ground condition, temperature and engine state. A two-sided dipstick may have different stopped/running scales; read the marking and manual. Clean the fill area and tools, avoid mixing incompatible fluids and do not overfill. Unexpected consumption or a repeated low level requires investigation, not endless topping up.

LEARNING STEP 5

Inspect and clean components as required

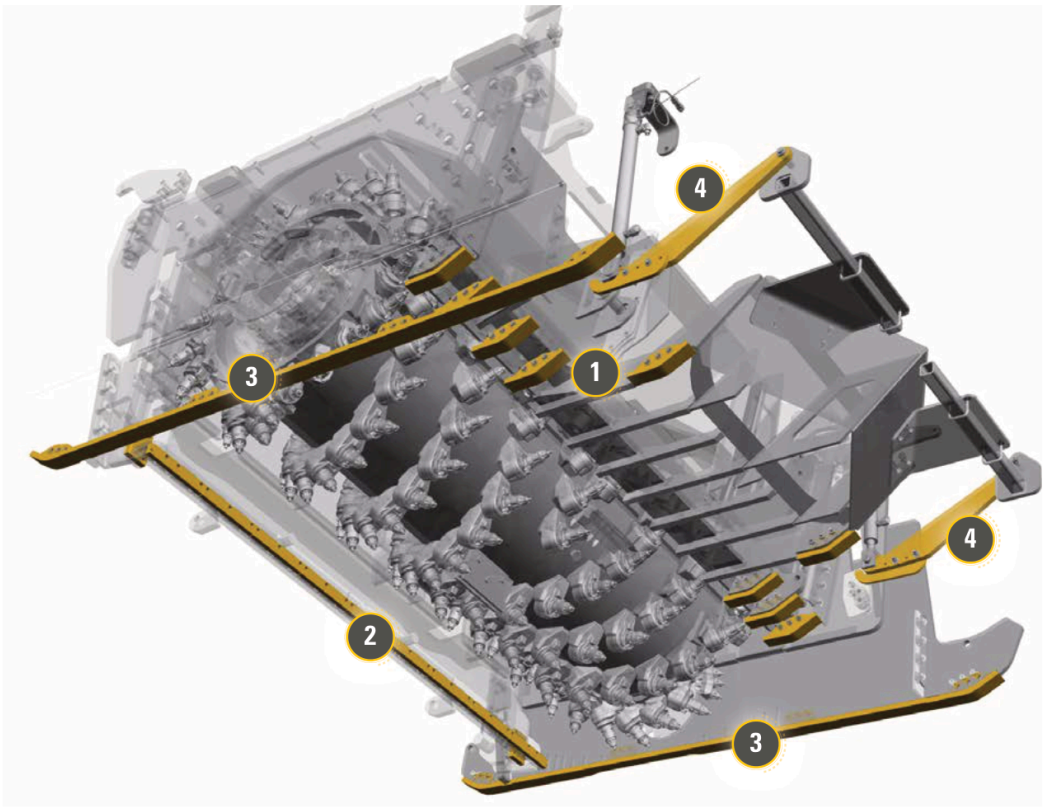
Inspect radiator and oil-cooler surfaces, plugs, dipsticks, battery condition, hoses and accessible guards. Cooling surfaces can be damaged by inappropriate pressure or tools. Hot coolant and batteries have their own hazards; use the safe procedure and authorized access. Record leak location and appearance without touching a pressurized leak. Electrical and mechanical repairs belong to authorized personnel.

LEARNING STEP 6

Housekeeping tasks

Remove rubbish and loose items from safe access areas. Clean excess grease, steps and handholds, and keep waste out of drains. Account for tools and restore guards before release. Enter meter reading, date, tasks, products, defects and next action in the maintenance record. A blank box is ambiguous: mark completed, not applicable with reason, or held for correction.

Four wear components around the cutting chamber



Note. Caterpillar. (2026). Cold Planer Parts Reference Guide (QEXQ3131-01 (05/2026), source PDF p. 18). Faithful crop; original labels retained. PM310, PM312, PM313, PM620, PM622, PM820, PM822 and PM825; individual parts/model tables apply. Full source context.

Worked example · The wrong grease assumption

A chart specifies a particular grease and service interval. A worker brings an unlabelled cartridge and proposes ten strokes at every fitting.

Think it through.

Do not use the unlabelled product or generic stroke count. Confirm lubricant, points, quantities and safe positions from the matching chart. Obtain the correct product and record any blocked or damaged fitting for action.